Q: Explain different Software Process Models

Ans:- → Software development lifecycle is a simplified representation of a software process. It consists of the planning models:-

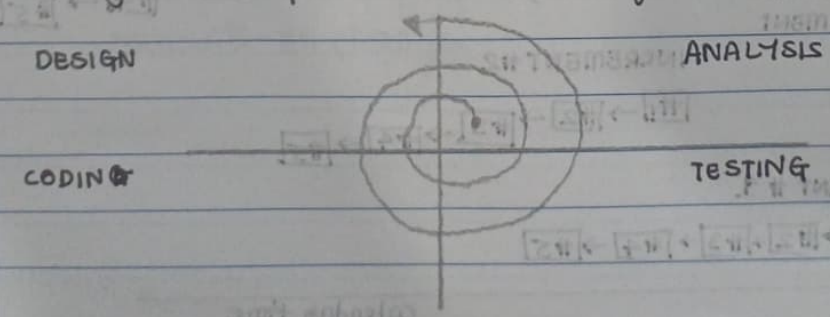
1) <u>The Waterfall Model</u>:- This is the classic, traditional model where development flows steadily downward through clearly defined phases. The phases must be completed and signed off before the next begins

COMMUNICATION → PLANNING → MODELLING → CONSTRUCTION
↓
DEPLOYMENT

• Advantages:- • Simple to understand, strict discipline, clear milestones

• Disadvantages:- • Inflexible to changes; working software is produced late, high risk if requirement are misunderstood

2) <u>The Spiral Model</u>:- It compines iterative development with systematic controls, moving in "loops" rather than a straight line. Its primary focus is risk analysis. In every loop you assess potential risk before writing code.

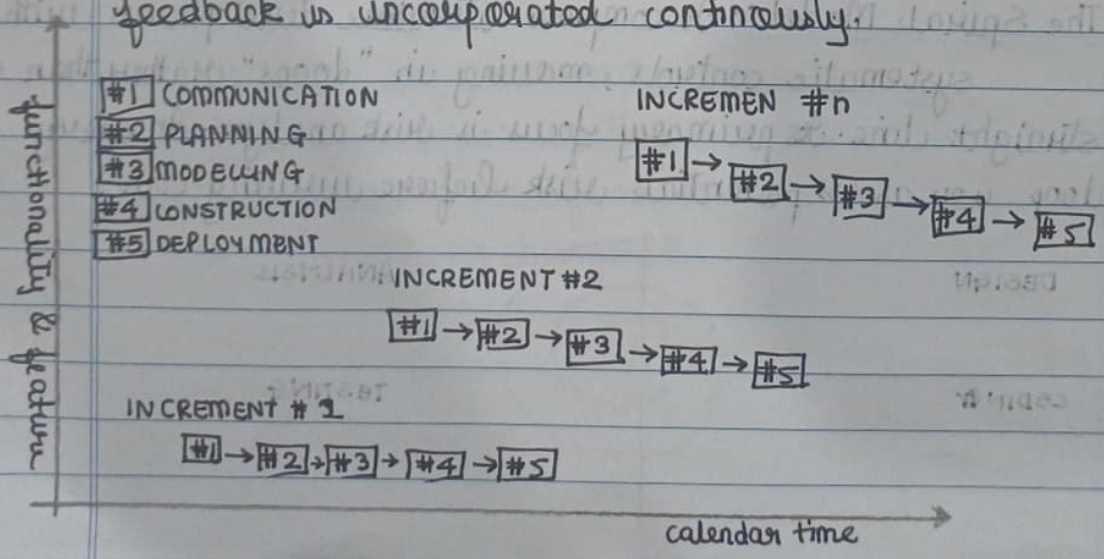


- Analysis:- determine objective, alternatives and constraint
- Design:- create a blueprint for the software architecture
- Coding:- develop & code the production for application
- Testing:- evaluates the software, deploy it & monitor it for maintenance

3) <u>The Incremental Model</u>:- This model is a software process model where the system is developed and delivered in small, manageable increment. Each increment adds new functioning to the previous release

- Phases in each increment:- 1] Requirement Analysis
  2] Design
  3] Implementation
  4] Testing
  5] Deployment

- In this software is delivered in parts; Each release is functional; Early delivery of working software; User feedback is incorporated continously.



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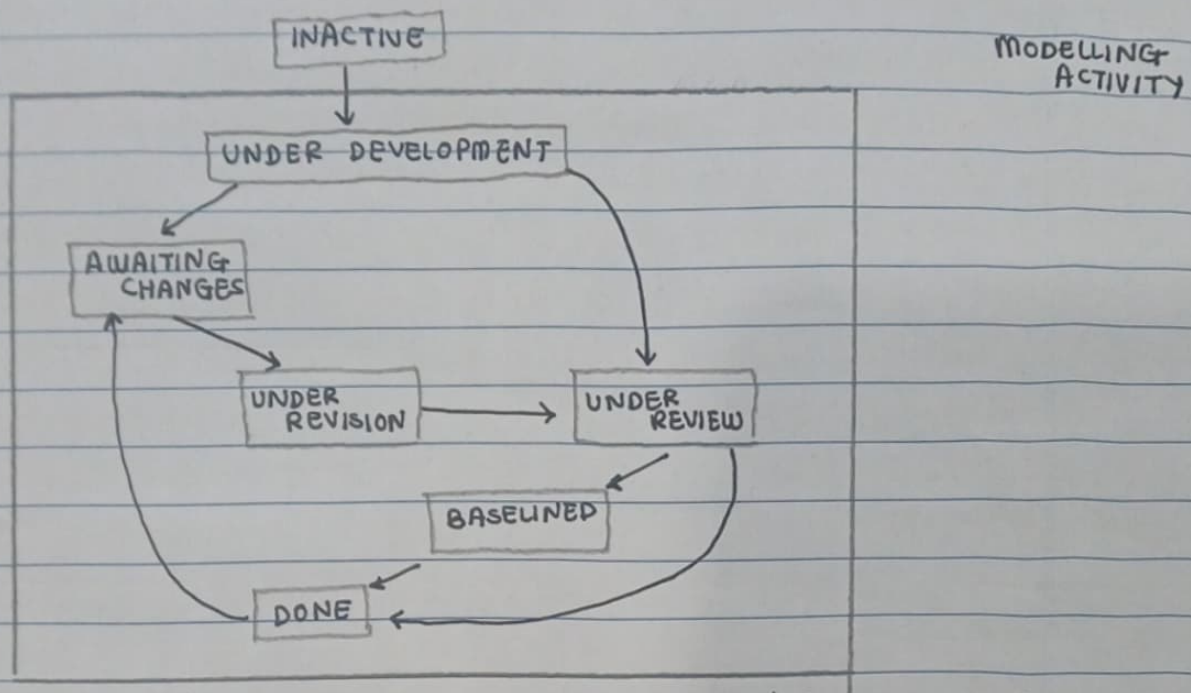
4] Concurrent Process Model:-



- This represents software development activities as states rather than rigid phases.
- Multiple activities processing simultaneously:

- States :- 1] Inactive :- activity hasn't started yet

2] Under development :- modelling work is currently created
3] Under review :- developed model is being reviewed
4] Baselined :- model is developed & placed under configuration control
5] Done :- activity is computed successfully
6] Awaiting changes :- waiting for changes or feedback
7] Under revision :- modification are being made our due to comments or changes.